Statistical Analysis - Practice Questions with Manual Calculations

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Q1. Five-Point Summary

Given the following dataset of house prices:

```python
prices = [120, 150, 180, 200, 210, 215, 220, 250, 300, 310]
```

(a) Explain what each component of the five-point summary represents.

(b) Compute the minimum, Q1, median, Q3, and maximum values using NumPy or Pandas.

(c) Interpret what this summary tells you about the data distribution.

Q2. Outlier Detection using IQR

You are given the following dataset of incomes:

```python
import pandas as pd
df = pd.DataFrame({'income':[22, 25, 29, 35, 120, 32, 30, 31, 28, 27]})
```

(a) Calculate Q1, Q3, and IQR manually using NumPy or Pandas.

(b) Determine the lower and upper bounds for outlier detection.

(c) Identify and remove all outliers from the dataset.

(d) Briefly explain when IQR is preferred over other methods.

Q3. Outlier Detection using Z-Score

```python
import pandas as pd
import numpy as np
df = pd.DataFrame({'age':[22, 23, 24, 25, 26, 27, 28, 55, 29, 30, 31]})
```

(a) Calculate the mean and standard deviation of the data.

(b) Compute the Z-score for each observation using the formula:

$$Z = \frac{X - \text{mean}}{\text{std}}$$

(c) Identify all values where |Z| > 3.

(d) Mention one limitation of the Z-score method for small datasets.

Q4. Hypothesis Testing – One-Sample Mean

```
python

scores = [85, 88, 90, 75, 95, 92, 87, 89, 91, 90]
```

(a) Assume the population mean is 85.

(b) Without using any statistical library, compute:

- The sample mean
- The standard deviation
- The t-statistic using:

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

(c) Discuss how you would decide whether to reject or fail to reject the null hypothesis if the significance level is 0.05.

Q5. Two-Sample Mean Comparison

```
python

groupA = [72, 75, 78, 79, 81, 76]
groupB = [68, 70, 74, 73, 75, 71]
```

(a) Compute the mean and standard deviation for both groups.

(b) Assuming unequal variances (Welch's t-test), calculate the t-statistic using only NumPy and manual computation.

(c) Explain the difference between equal variance and unequal variance assumptions.

(d) Interpret your findings logically without referring to any library function.

Q6. ANOVA Test (Manual Approach)

Three fertilizers (A, B, and C) were tested on crop yield:

```python
A = [12, 15, 14, 10, 13]
B = [9, 8, 10, 9, 11]
C = [14, 16, 15, 13, 17]
```

(a) Calculate the overall mean.

(b) Compute Between-Group Sum of Squares (SSB) and Within-Group Sum of Squares (SSW).

(c) Derive the F-ratio manually:

$$F = \frac{\text{MSB}}{\text{MSW}}$$

(d) Explain what a high F-value indicates in this context.

Q7. Chi-Square Test of Independence

Gender	Likes Coffee	Likes Tea
Male	30	20
Female	10	40

(a) Write down the observed frequency table as a 2D NumPy array.

(b) Compute the row totals, column totals, and expected frequencies manually using:

$$E = \frac{(\text{Row Total}) \times (\text{Column Total})}{\text{Grand Total}}$$

(c) Calculate the Chi-square statistic using:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

(d) Explain whether the variables appear independent based on your computed statistic.

Q8. Correlation Analysis

```python
import pandas as pd
df = pd.DataFrame({
   'temperature': [20, 22, 25, 27, 30, 32, 35, 37],
   'sales': [200, 220, 250, 270, 300, 320, 350, 370]
})
```

(a) Compute the mean of both columns.

(b) Calculate the Pearson correlation coefficient (r) manually using:

$$r = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sqrt{\sum(x - \bar{x})^2 \sum(y - \bar{y})^2}}$$

(c) Interpret whether the relationship is positive, negative, or negligible.

Q9. Percentiles and Quartiles

```python
data = [10, 15, 20, 25, 30, 35, 40, 45, 50]
```

(a) Explain the difference between percentiles and quartiles.

(b) Manually compute the 25th, 50th, and 90th percentiles using NumPy's percentile function.

(c) Verify the results using the mathematical percentile formula (without built-in methods).

Q10. Mini Project – Statistical Exploration

You are given a dataset `students.csv` with columns: **score**, **study_hours**, and **gender**.

Perform the following:

1. Compute the five-point summary for `score`.
2. Identify and treat outliers using IQR or Z-score.
3. Compare mean scores between genders using a manual two-sample test.
4. Compute and visualize the correlation between `score` and `study_hours` using Pandas and Matplotlib.
5. Write short interpretations for each step.

📚 Key Formulas Reference

Descriptive Statistics

- **Mean**: $\bar{x} = \frac{\sum x_i}{n}$
- **Variance**: $s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$
- **Standard Deviation**: $s = \sqrt{s^2}$

Outlier Detection

- **IQR**: $\text{IQR} = Q_3 - Q_1$
- **Lower Bound**: $Q_1 - 1.5 \times \text{IQR}$
- **Upper Bound**: $Q_3 + 1.5 \times \text{IQR}$

Hypothesis Testing

- **One-Sample t-test**: $t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$
- **Welch's t-test**: $t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$

ANOVA

- **SSB**: $\sum n_i (\bar{x}_i - \bar{x})^2$
- **SSW**: $\sum \sum (x_{ij} - \bar{x}_i)^2$
- **MSB**: $\frac{\text{SSB}}{k-1}$

- **MSW**: $\frac{\text{SSW}}{N-k}$

💡 Learning Objectives

By completing these questions, you will:

✅ Understand descriptive statistics beyond library functions

✅ Master outlier detection techniques (IQR and Z-score)

✅ Perform hypothesis tests with manual calculations

✅ Comprehend the mathematical foundations of statistical tests

✅ Apply statistical methods to real-world datasets

🎯 Tips for Success

1. **Show all work -** Write out each step of your calculations
2. **Use NumPy for basic operations** (mean, std) but calculate statistics manually
3. **Verify results -** Compare manual calculations with library functions
4. **Interpret meaningfully -** Don't just compute; explain what the numbers mean
5. **Practice formula manipulation -** Understanding formulas is key to mastery

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Good Luck! 🚀